AN IMMERSION WITH QUASI-COMPACT SOURCE OR REDUCED SOURCE IS AN OPEN IMMERSION ONTO A DENSE OPEN OF ITS SCHEME THEORETIC IMAGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement carries the disjunctive hypothesis (quasi-compact or reduced) and an 'Moreover' clause on topological density; both are inside the counted statement span.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Commutation of the scheme theoretic image with etale localization in the quasi-compact case. Lemma lemma-quasi-compact-scheme-theoretic-image (3249-3265), used at proof lines 3580-3582. Handles the first alternative of the hypothesis.
- The scheme theoretic image of a morphism from a reduced space. Lemma lemma-scheme-theoretic-image-reduced (3286-3296), used at proof lines 3582-3583. Handles the second alternative of the hypothesis.
- The scheme case for a quasi-compact immersion. Morphisms, Lemma morphisms-lemma-quasi-compact-immersion, used at proof lines 3584-3585. Is the input that closes the argument.

Source reference. Original lemma 082Z: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10269–10269. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0830: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10270–10270. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01RG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2301–2301. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let S be a scheme. Let $h : Z \to X$ be an immersion of algebraic spaces over S. Assume either $Z \to X$ is quasi-compact or Z is reduced. Let $\overline{Z} \subset X$ be the scheme theoretic image of h. Then the morphism $Z \to \overline{Z}$ is an open immersion which identifies Z with a scheme theoretically dense open subspace of $\overline{Z}$. Moreover, Z is topologically dense in $\overline{Z}$.*

Proof. In both cases the formation of $\overline{Z}$ commutes with étale localization, see Lemmas 082Z and 0830. Hence this lemma follows from the case of schemes, see Morphisms, Lemma 01RG. □